CE205 Databases and Information Retrieval

Lab 1

(Week 3)

Marcia's Dry Cleaning

SQL Queries

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/*****   Case Question  A   ************************************************/

/*****   CUSTOMER DATA  ***************************************************/

SELECT * FROM CUSTOMER;

/*****   INVOICE  *********************************************************/

SELECT * FROM INVOICE;

/*****   INVOICE_ITEM  ****************************************************/

SELECT * FROM INVOICE_ITEM;



/*****   Case Question  B   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER;



/*****   Case Question  C   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER

WHERE  FirstName = 'Nikki';



/*****   Case Question  D   ************************************************/

SELECT LastName, FirstName, Phone, DateIn, DateOut

FROM   CUSTOMER, INVOICE

WHERE  CustomerID = CustomerNumber

AND    TotalAmount > 100;
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/*****   Case Question  E   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER

WHERE  FirstName Like 'B%';



/*****   Case Question  F   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER

WHERE  LastName Like '%cat%';



/*****   Case Question  G   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER

WHERE  PHONE LIKE '_23%';

/*****   Case Question  H   ************************************************/

SELECT Max(TotalAmount), Min(TotalAmount)

FROM INVOICE;



/*****   Case Question  I   ************************************************/

SELECT Avg(TotalAmount)

FROM INVOICE;



/*****   Case Question  J   ************************************************/

SELECT Count(CustomerID)

FROM   CUSTOMER;
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/*****   Case Question  K   ************************************************/

SELECT LastName, FirstName

FROM CUSTOMER

GROUP BY LastName, FirstName;



/*****   Case Question  L   ************************************************/

SELECT LastName, FirstName, COUNT(*) AS Last_First_Combination_Count

FROM CUSTOMER

GROUP BY LastName, FirstName;



/*****   Case Question  M   ************************************************/

SELECT LastName, FirstName, Phone

FROM CUSTOMER

WHERE CustomerID IN

                    (SELECT CustomerNumber

                    FROM INVOICE

                    WHERE TotalAmount > 100)

ORDER BY LastName, FirstName DESC;



/*****   Case Question  N   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER, INVOICE

WHERE  CustomerID=CustomerNumber And TotalAmount>100

ORDER BY LastName, FirstName DESC;
```

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/*****   Case Question  O   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER JOIN INVOICE ON

      CustomerID=CustomerNumber

WHERE  TotalAmount>100

ORDER BY LastName, FirstName DESC;



/*****   Case Question  P   ************************************************/

SELECT LastName, FirstName, Phone

FROM CUSTOMER

WHERE CustomerID IN (SELECT            CustomerNumber

                              FROM     INVOICE

                              WHERE    InvoiceNumber IN

                                             (SELECT InvoiceNumber

                                             FROM INVOICE_ITEM

                                             WHERE Item = 'Dress Shirt'))

ORDER BY LastName, FirstName DESC;



/*****   Case Question  Q   ************************************************/

SELECT LastName, FirstName, Phone

FROM   CUSTOMER, INVOICE, INVOICE_ITEM

WHERE  CustomerID=CustomerNumber

AND    INVOICE.InvoiceNumber=INVOICE_ITEM.InvoiceNumber

AND    Item='Dress Shirt'

ORDER BY LastName, FirstName DESC;
```

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/*****   Case Question  R   ************************************************/

SELECT FirstName, LastName, Phone

FROM   CUSTOMER JOIN INVOICE ON

CustomerID=CustomerNumber

JOIN INVOICE_ITEM ON INVOICE.InvoiceNumber=INVOICE_ITEM.InvoiceNumber

WHERE  Item='Dress Shirt'

ORDER BY LastName, FirstName DESC;

/*****   Case Question  S   ************************************************/

SELECT LastName, FirstName, Phone

FROM CUSTOMER JOIN INVOICE

ON CustomerID = CustomerNumber

WHERE      InvoiceNumber IN

                        (SELECT InvoiceNumber

                         FROM INVOICE_ITEM

            WHERE Item = 'Dress Shirt')

ORDER BY LastName, FirstName DESC;

/* Alternate version, with join in subquery: */

SELECT LastName, FirstName, Phone

FROM CUSTOMER

WHERE CustomerID IN

(SELECT CustomerNumber

FROM INVOICE JOIN INVOICE_ITEM

ON INVOICE.InvoiceNumber = INVOICE_ITEM.InvoiceNumber

                                    WHERE Item = 'Dress Shirt')

ORDER BY LastName, FirstName DESC;
```

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/*****   Case Question  T   ************************************************/

/*  LEFT OUTER JOIN SOLUTION  */

SELECT LastName, FirstName, Phone, TotalAmount

FROM CUSTOMER C LEFT JOIN (INVOICE I JOIN INVOICE_ITEM II

        ON I.InvoiceNumber = II.InvoiceNumber AND II.Item = 'Dress Shirt')

        ON C.CustomerID = I.CustomerNumber

ORDER BY TotalAmount, LastName, FirstName DESC;



/*  UNION SOLUTION  */

SELECT LastName, FirstName, Phone, TotalAmount

FROM CUSTOMER C, INVOICE I, INVOICE_ITEM II

WHERE C.CustomerID = I.CustomerNumber AND I.InvoiceNumber =

        II.InvoiceNumber AND II.Item = 'Dress Shirt'

UNION

SELECT LastName, FirstName, Phone, NULL

FROM CUSTOMER

WHERE CustomerID NOT IN

        (SELECT CustomerNumber

         FROM INVOICE I, INVOICE_ITEM II

         WHERE I.InvoiceNumber = II.InvoiceNumber AND II.Item = 'Dress Shirt')

ORDER BY TotalAmount, LastName, FirstName DESC;



/*  RIGHT OUTER JOIN SOLUTION  */

SELECT LastName, FirstName, Phone, TotalAmount

FROM (INVOICE I JOIN INVOICE_ITEM II
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```
        ON I.InvoiceNumber = II.InvoiceNumber AND II.Item = 'Dress Shirt') RIGHT JOIN CUSTOMER C

        ON C.CustomerID = I.CustomerNumber

ORDER BY TotalAmount, LastName, FirstName DESC;
```